Report Title: The Impact of Video Game Experience on Response Time: A Go/No-Go Task

Abstract

Video Gaming is a widespread activity, and growing evidence suggests it may influence cognitive functions such as attention and response inhibition. This study aimed to investigate whether different levels of gaming experience (non-gamers, casual gamers, experienced gamers) are associated with differences in response time on a Go/No-Go task. Fifty-five participants took part and were placed into one of the three groups based on how much they play games each week. They then completed a computerised Go/No-Go task that recorded their reaction times. Although descriptive statistics suggested that experienced gamers responded slightly faster than non-gamers, the one-way ANOVA showed no significant differences between groups. These results suggest that gaming experience may not be reliability linked to improved reaction times, though further research is needed to clarify this relationship.

Introduction

Video games are a central part of modern leisure activity, with millions of people worldwide engaging in gaming regularly. With more people playing games, researchers have started to look at whether gaming might actually influence how we think and process information. In particular, researchers have explored whether video game play enhances reaction times, a cognitive skill associated with attention, decision making, and information speed processing (Granic et al., 2014). Reaction time, often measured with tasks like the Go/No-Go, reflects how quickly one responds to stimuli (Wessel, 2018).

Previous research suggests that action video games can enhance perception, attention, and reaction time. Studies by Green and Bavelier (2013) and Dye et al. (2009) found gamers performed better on visual tasks and processing speed, supporting the idea that gaming may offer cognitive benefits, particularly in fast-response situations.

However, the findings are mixed. Some research suggests cognitive benefits may be limited to certain game types or populations. Bediou et al. (2018) found that while action games can boost specific skills, the effects aren't universal. Boot et al. (2011) also argues that factors like motivation, expectations, or self-selection may explain cognitive differences, rather than gaming itself. These issues reflect broader concerns about flexible methods and false positives in psychology (Simmons, Nelson, & Simonsohn, 2011).

Many earlier studies have simply compared gamers to non-gamers, treating gaming experience as a black-and-white category. But in reality, gaming habits vary widely. Someone who plays occasionally for fun may not show the same cognitive patterns as someone who games regularly, especially with fast-paced, reaction-heavy games. By dividing participants into three more realistic groups- non-gamers, casual gamers, and experienced gamers- this study takes a more detailed approach. It allows for a better understanding of how different levels of gaming experience might relate to reaction time.

This study examined whether gaming experience relates to reaction time using a Go/No-Go task. Unlike past research focused on expert or action gamers, it used broader categories to capture general gaming experience. It was expected that experienced gamers would respond fastest, non-gamers slowest, and casual gamers would fall in between, based on prior links between gaming and attentional processing.

By including three distinct experienced levels, this study seeks to fill a gap in the literature and contribute to a more nuanced understanding of the relationship between video gaming

and cognitive function- particularly in areas such as attentional control and response inhibition, which continue to develop during early adulthood (Dumontheil, 2016).

Method

Design

A between-groups quasi-experimental design was employed. The independent variable (IV) was video game experience, with three levels: non-gamers (0 hours/week), casual gamers (<8 hours/week), and experienced gamers (>8 hours/week). The dependent variable was participants' mean reaction time (in milliseconds) measured using a Go/No-Go task.

Participants

Fifty-five participants were recruited using snowball and opportunity sampling. The sample consisted of 29 biological females (M age= 25, SD = 9, age range = 18-50), 23 biological males (M age = 26, SD = 12, age range = 18-54), and 3 participants who preferred not to disclose their gender (M age = 20, SD = 3, age range = 18-24). Participants were categorised into three groups based on their gaming experience: 19 non-gamers, 19 casual gamers, and 17 experienced gamers.

Materials

Participants used a laptop or PC with a physical keyboard to complete a Go/No-Go task via Gorilla Experiment Builder. Stimuli included the letters "P" (Go) and "R" (No-Go), each displayed over a yellow star in a 2x2 grid. Letters appeared every 500ms, with a 1500ms response window. Participants completed 72 trials. The task was chosen for its high test-retest reliability, ease of use, and strong foundation in past research on response inhibition.

Procedure

Participants first viewed an information page outlining the study, followed by a consent form. After providing consent, they completed a demographic questionnaire and selected their gaming experience level. They then completed the Go/No-Go task. Finally, a debrief page was presented at the end of the study.

Results

The average reaction times and standard deviations for each group are shown in Table 1.

Table 1

Mean Reaction Times by Gaming Group

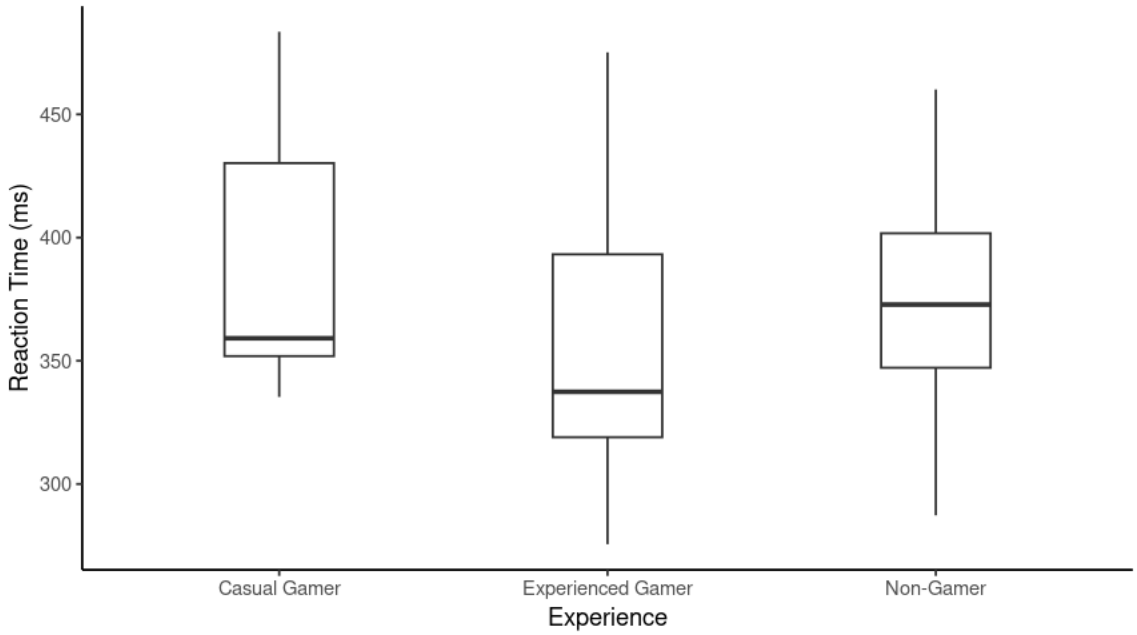
Group	N	Mean RT (ms)	SD
Non-Gamers	19	375	51.2
Casual Gamers	19	390	47.3
Experienced Gamers	17	354	54.3

A one-way ANOVA was conducted to compare the mean reaction times across the three gaming experience groups. The analysis revealed there was no statistically significant difference: $F(2, 52) = 2.20$, $p = 0.12$.

Figure 1

A box plot showing mean reaction times by group

(See Appendix for R output and figure)



While experienced gamers had a lower mean reaction time than casual and non-gamers, the interquartile ranges showed considerable overlap. The standard deviations were also quite similar across all three groups, suggesting that differences in reaction speed time were not consistent enough to reach statistical difference. Together, these patterns reflect the ANOVA finding that gaming experience did not significantly affect reaction time.

Discussion

This study explored whether different levels of video game experience are associated with variations in reaction time, using a Go/No-Go task. It was hypothesised that experienced gamers would respond more quickly than non-gamers, with casual gamers falling in between. However, the findings did not support this hypothesis. The one-way ANOVA showed no statistically significant differences in reaction times across the three groups. However, descriptive trends in the data hinted that experienced gamers might have responded slightly faster on average that non-gamers, whereas casual gamers appeared to have the slowest reaction times.

These findings contrast with previous research suggesting cognitive advantages linked to video gaming. For instance, Green and Bavelier (2013) reported that action gamers show enhanced attentional control and quicker visual processing. Similarly, Dye et al. (2009) found improvements in processing speed across different age groups. The lack of significant results in the current study may come from several factors, such as limitations in task design, a small sample size, or uncontrolled variables like specific game genres participants played.

A key limitation is how gaming experience was defined. While using three gamer categories added nuance, it didn't account for game type. Research has highlighted that action games in particular are more likely to influence cognitive functions like attention and perception (Bediou et al., 2018), whereas narrative or casual games may have little to no impact. By not accounting for game type, our categorisation may have masked potential group differences.

Another limitation is the reliance on self-reported gaming experience. Despite clear group definitions—non-gamers, casual gamers, experienced gamers—accurate self-assessment was still required, leaving room for recall bias or social desirability effects. Boot et al. (2011) also

noted that observed cognitive benefits in gamers may stem from self-selection or placebo effects. This ties into broader methodological issues in psychology, such as flexible data analysis and lack of transparency, which can inflate false-positive rates (Simmons, Nelson, & Simonsohn, 2011).

The Go/No-Go task itself, while widely used for assessing response inhibition (Wessel, 2018), may have lacked the sensitivity needed to detect subtle differences between groups. Wiemers and Redick (2019) emphasise that features such as timing and task complexity can heavily influence performance on such measures. It's possible that the task used was not demanding enough to reveal any performance advantage associated with gaming experience.

Age may have also played a role. The sample largely comprised young adults- a group typically at their peak in terms of cognitive control (Dumontheil, 2016). Gaming-related cognitive effects might be more noticeable in groups with greater variability in these abilities, such as children or older adults. Future research could explore whether gaming's cognitive impact varies by age, potentially revealing when video games are most beneficial.

Even though the main hypothesis was not supported, this study adds to the literature by taking more of a refined approach to categorising gaming experience. It also highlights the need for future research to move beyond simple gamer/non-gamer distinctions and instead look at more detailed behavioural data- such as specific game types. Longitudinal studies may also help clarify whether games cause cognitive improvements or if pre-existing cognitive traits lead individuals to engage in gaming.

Overall, while no statistically significant groups differences in reaction time were found, these findings highlight the complexity of how video game experience relates to cognitive performance. As research progresses, using more detailed methods and considering individual differences, like age, will be crucial to understanding how gaming affects cognition.

References

Bediou, B., Adams, D. M., Mayer, R. E., Tipton, E., Green, C. S., & Bavelier, D. (2018). Meta-analysis of action video game impact on perceptual, attentional, and cognitive skills. Psychological Bulletin, 144(1), 77–110. https://doi.org/10.1037/bul0000130

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Dumontheil, I. (2016). Adolescent brain development. Current Opinion in Behavioral Sciences, 10, 39–44. https://doi.org/10.1016/j.cobeha.2016.04.012

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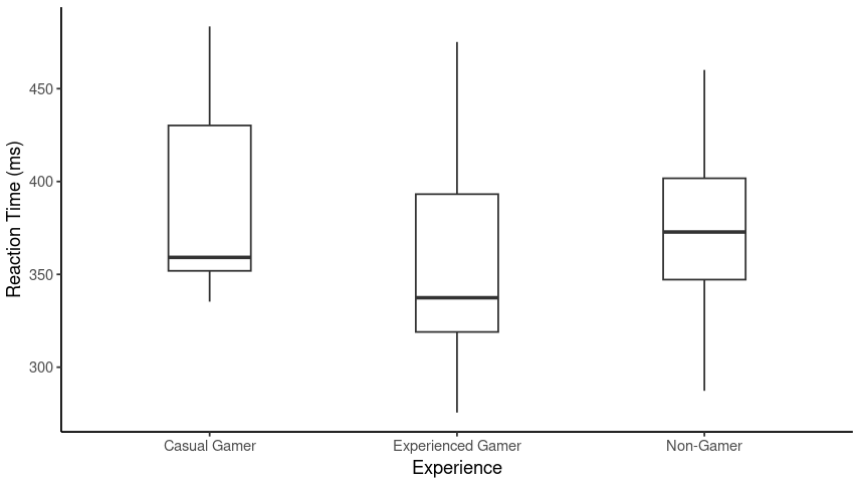
Wiemers, E. A., & Redick, T. S. (2019). Working memory capacity and intraindividual variability of proactive control. Acta Psychologica, 198, 102868. https://doi.org/10.1016/j.actpsy.2019.102868

Appendix (R Code and R Output-not needed if you are a Law Student)

R Output from One-Way ANOVA

```
R 4.4.2 · /cloud/project/
> describe_across(data = final,
+                 by = experience,
+                 variable = go_reaction_time,
+                 functions = list(mean = mean, sd = sd, med = median, iqr = IQR),
+                 pivot = TRUE)
# A tibble: 3 × 6
  experience        variable          mean    sd   med   iqr
  <chr>             <chr>            <dbl> <dbl> <dbl> <dbl>
1 Casual Gamer      go_reaction_time  390.  47.3  359.  78.3
2 Experienced Gamer go_reaction_time  354.  54.3  337.  74.2
3 Non-Gamer         go_reaction_time  375.  51.2  373.  54.6
> summary(m1)
            Df Sum Sq Mean Sq F value Pr(>F)
experience   2  11413    5706   2.203  0.121
Residuals   52 134686    2590
>
```

Boxplot of Reaction Times by Gaming Experience Group



Ethical Considerations

This study followed the British Psychological Society (BPS) ethical guidelines, with particular care taken to ensure participants were treated respectfully and responsibly. Before starting, all participants read an information sheet that clearly explained the study's aims, what they would be asked to do, and their rights—including the choice to withdraw at any time without giving a reason. Informed consent was then obtained digitally.

The Go/No-Go task used was low-risk and designed to be short to reduce the chance of fatigue. Participants who took too long to complete the task were excluded to protect data quality and avoid unnecessary strain. No post-hoc decisions were made, helping to support transparency.

In line with open science principles, the study was pre-registered, and key elements such as the task, sampling method, inclusion criteria, and analysis plan were all confirmed in advance. Data were anonymised and stored in a shareable CSV file, making it ready for public access while protecting privacy. RStudio (an open-source tool) was used for data analysis to support reproducibility. Overall, ethical, and open research practices were central to the study's design, going beyond basic requirements.

Grading Matrix

Note: Criteria are equally weighted (i.e., all criteria impact upon your grade to an equal amount)

Criteria	1st	2.1	2.2	3rd	Marginal Fail	Fail
1 Understanding and evaluating previous literature and formulating a research problem (Introduction)	The report contains evidence of **breadth and depth of reading** to inform development of the psychological study. The student demonstrates **excellent knowledge and understanding** of the topic under investigation **where there is evidence of key literature and psychological theory/explanation being explored in detail and there is some evidence of an analytical approach**. The student clearly states the research aims and hypotheses, and these follow **convincingly** from the review of the literature.	The report contains a review of the literature that provides the readers the **background needed** to understand the context for the psychological study. The student demonstrates **very good knowledge and understanding where there is some depth of exploring psychological literature or psychological theory/explanation, but this may not be focussed or done in much depth, perhaps with no analytical approach taken. The student clearly** states the research aims and hypotheses, and these follow **logically** from the review of the literature.	The report contains a review of the literature that provides the readers with **some of the background** needed to understand the context for the psychological study. The student attempts to demonstrate **good knowledge and understanding** of the topic under investigation **but this is evidenced through a more descriptive understanding of studies and theory where there may be inaccuracies, a focus on historic literature or lack of depth.** Aims and hypotheses may be stated, but these may **not lead logically** from the literature that has been reviewed.	The report contains a **sparse review of the literature** that provides the readers with **only limited background** needed to understand the context for the psychological study. The student demonstrates **adequate knowledge and understanding** of the topic under investigation **but there is only a limited surface-level exploration of psychological literature with little-to-no psychological theory or explanations discussed.** Student makes some attempt at stating the research aims and question, but these may be **unclear/may not lead logically from the literature reviewed**.	The report **contains minimal literature** and insufficient background to understand the context for the psychological study. There are no/or poor arguments presented to motivate the research. The report is likely either **lacking any research aims or hypotheses**, or those that are presented are very unclear.	The report contains **negligible literature**. **No** arguments are made to motivate the research. **No** research aims or hypotheses are presented.

2 Understanding of the formal reporting of methods and design (Method)	Excellent knowledge and understanding of the methods and design appropriate to the topic under investigation and its underlying concepts **where the Method section is written clearly with sufficient details provided to replicate the study precisely,** and there is a **clear understanding of more nuanced psychological research methods** such as, but not limited to, extraneous variables.	Very good knowledge and understanding of the methods and design appropriate to the topic under investigation and its underlying concepts. **The Method section is written with sufficient details given to replicate study, but there may be evidence of either slight misunderstandings, misreporting or missing minor elements to precisely replicate the study,** or **no information to demonstrate more nuanced understanding of psychological research methods.**	Generally good knowledge and understanding of the methods and design appropriate to the topic under investigation and its underlying concepts. **The Method section is written with most of the information required for replication, but with some misunderstanding of terminology and/or missing key methodological details.**	Sufficient knowledge and understanding shown of the methods and design appropriate to the topic under investigation and its underlying concepts. **The Method section is written reasonably, in which some details given to allow replication, but with important and necessary details missing that would be needed to replicate the study well.**	Insufficient knowledge and understanding shown of the methods and design appropriate to the topic under investigation and its underlying concepts. **Poor writing of the Method section; Details given would allow only a partial replication of the study.**	Highly insufficient knowledge and understanding shown of the methods and design appropriate to the topic under investigation and its underlying concepts. **Insufficient writing of the Method section; Replication of the study would not be possible on the basis of the methods provided. Highly insufficient demonstration of relevant skills** (in relation to writing a Method section).
3 Understanding of the data that were gathered and how to report them formally (Results)	Excellent understanding of the data and the results of inferential tests where **all results have been communicated correctly and clearly, including interpretative discussion of descriptive statistics and these are visualised appropriately.**	Very good understanding of the data and the results of inferential **tests where results have been communicated mostly correctly, but may be some slight inaccuracies or missing information, possibly within the visualisation of the data.**	Generally good understanding of the data and the results of inferential tests but **there may be elements mis-reported or missing information, possibly within, but not limited to, within the discussion of descriptive statistics or the data visualisation.**	Sufficient understanding of the data and the results of inferential tests in which information concerning descriptive statistics and inferential statistics are provided, **but with fundamental flaws present** (e.g. incorrect conclusions regarding significance). **Data visualisation may not be presented correctly.**	Insufficient understanding of the data and the results of inferential tests. **The appropriate descriptive and/or inferential test information are not reported or are serious flaws in understanding. Data visualisation may also be absent or does not aid the reader at all**.	Highly insufficient understanding of the data/results of inferential tests where **appropriate important information concerning inferential and descriptive statistics is not included. Highly insufficient demonstration of relevant skills** (in relation to writing a Results section).

4 Understanding findings and drawing conclusions (Discussion)	The report contains an accurate and clear interpretation of the study's main findings with **insightful discussion of findings (e.g. offering theoretical explanation) in light of previous theory and goes beyond generically stating whether the results conform or not to previous literature. There is evidence of an analytical approach taken in discussing the methods, limitations and future direction of the work.**	The report contains an accurate interpretation of the study's main findings. **There are good attempts at discussing the findings in light of previous literature, but there is scope for further depth when offering theoretical explanations, or there may not be an overly analytical approach taken when discussing the methods, limitations, and future direction of the work.**	The report contains a generally accurate attempt to interpret the study's main findings. There is some discussion of the findings in light of previous literature and theory, **but more insightful explanations could have been provided, and there is reliance on simply stating whether results conform or not to previous results. There is an attempt at considering the implications and limitations of the work, but this tends to consist of the 'generic issues' that students usually consider (e.g. sample size).**	The student makes an attempt to interpret the study's findings, but these interpretations may **be incorrect/not based on evidence.** There is **limited discussion of the findings in light of previous literature and theory. There is minimal analysis of the implications of work or the methods.**	The student makes an attempt to interpret the study's findings, but these interpretations are incorrect/not based on evidence. **There is little to no discussion** of the findings in light of previous literature and theory. There **is almost no consideration** of the implications of work or limitations in the methods.	**Highly insufficient or no attempt** to interpret main findings. **Highly insufficient or no consideration** of the results in light of previous literature/theory. There is **no consideration** of the implications of work or limitations in the methods.
5 Communication and presentation (Overall)	Excellent communication skills; The report is written with **structure and language that means the student's intended meaning is quick and easy to determine, and concepts are explained precisely, carefully, and concisely.** Referencing is **complete and accurate throughout** using APA or Harvard style.	Very good communication skills; The report is written with **structure and language that means the student's intended meaning is usually quick and easy to determine.** Referencing **is generally accurate** using APA or Harvard style.	Good communication skills; The report is written with structure and language that means it **is normally possible to determine the student's intended meaning, albeit with significant effort in places.** Good attempt at referencing using APA or Harvard style, but **may have some errors or omissions.**	Generally competent communication skills; The report is written with structure and language that means it is **normally possible to determine the student's intended meaning, albeit with more effort in places.** Some attempt made at referencing, but **no identifiable referencing style is used consistently.**	Limited clarity in communication skills; The writing and structure of the report makes it **very difficult to determine the student's intended meaning.** Report writing guidelines may not have been followed, and the structure is not coherent. **Very limited** attempt at referencing.	**Fails** to address most or all of the requirements of the brief. **Fails** to demonstrate relevant skills. Lacks basic communication skills. **Almost no** attempt at referencing.

6. Ethical Considerations	**Excellent consideration** of the ethical issues, and **coherent discussion** of how they were addressed when designing the study **where the points raised have been analysed more thoroughly or that the student has demonstrated thinking about more nuanced ethical elements beyond more the generic ethical issues that occur in most experimental paradigms.**	**Very good consideration** of the ethical issues, and **clear discussion** of how they were addressed when designing the study. **Though the discussion is not overly analytical in nature, or the points raised may not go beyond more generic ethical issues that were raised**.	**Generally good consideration** of the ethical issues, and **some discussion** of how they were addressed when designing the study. **Possibly some omissions and/or weaknesses, or the points don't go too far beyond more generic ethical issues in experimental paradigms.**	**Some consideration** of the ethical issues, but **little attempt at discussing** of how they were addressed when designing the study.	**Limited consideration** of the ethical issues, and **poor/absent discussio**n of how they were addressed in designing the study.	**No ethical issues discussed. Section missing.**
7. Engagement with collaborative practice and engagement with lab session activities	**100% to 90%** attendance and engagement. Attendance of the term 2 lab sessions (percentage attended) and engagement with the term 2 lab session activities (percentage completed) are equally weighted to calculate the overall percentage.	**89% - 75%** attendance & engagement. Attendance of the term 2 lab sessions (percentage attended) and engagement with the term 2 lab session activities (percentage completed) are equally weighted to calculate the overall percentage.	**74% to 60%** attendance and engagement. Attendance of the term 2 lab sessions (percentage attended) and engagement with the term 2 lab session activities (percentage completed) are equally weighted to calculate the overall percentage.	**59% - 40%** attendance & engagement. Attendance of the term 2 lab sessions (percentage attended) and engagement with the term 2 lab session activities (percentage completed) are equally weighted to calculate the overall percentage.	**39% - 20%** attendance & engagement. Attendance of the term 2 lab sessions (percentage attended) and engagement with the term 2 lab session activities (percentage completed) are equally weighted to calculate the overall percentage.	**Less than 20%** attendance & engagement. Attendance of the term 2 lab sessions (percentage attended) and engagement with the term 2 lab session activities (percentage completed) are equally weighted to calculate the overall percentage.

NB: An exceptional first MAY be awarded if all criteria are graded at first class level